

Nayeon Kim

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Work Experience

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- Samsung Electronics AI Center, Korea - AI Researcher** *Mar 2019 – Present*
- Developed a generative model for industrial process automation by predicting time-series data.
 - Developed computer vision models for high-level semantic and geometric understanding of road scenes
: Online HD Map construction (NeurIPS '24), 3D lane detection (ICASSP '23)
- Korea Electronics Technology Institute (KETI), Korea - Research Intern** *Dec 2017 – Oct 2018*
- Developed a computer vision model for monocular 3D perception: Monocular depth prediction (KSAE '18)

Publications

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- Unveiling the Hidden: Online Vectorized HD Map Construction with Clip-Level Token Interaction and Propagation** *NeurIPS 2024*
Nayeon Kim*, Hongje Seong*, Daehyun Ji, Sujin Jang
- D3DLD: Depth-aware Voxel Space Mapping for Monocular 3D Lane Detection with Uncertainty** *ICASSP 2023*
Nayeon Kim*, Moonsub Byeon*, Daehyun Ji, Dokwan Oh
- Unsupervised Depth Prediction and Camera Motion Estimation in a Dynamic Environment** *KSAE 2018*
Nayeon Kim, MinGyu Park, Youngbae Hwang

Education

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- Kookmin University (KMU)** *Mar 2015 – Feb 2019*
B.S. in Computer Science, Automotive Software Design
- GPA: 4.19/4.5 (**Rank : 1/49**)

Patents

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- Method for Automated Analyzing Semiconductor Samples Using Future Frame Generation** *In Progress*
Nayeon Kim, Jiwon Lee, Zyup Jeong, Sujin Jang
- Method for Automated Preparation of Semiconductor Analysis Samples Using a Self-Learning AI Decision Model** *In Progress*
Jiwon Lee, Nayeon Kim, Zyup Jeong, Sujin Jang
- Method and Apparatus with Vehicle Driving Control** *US-Application No. 18/936,286*
Hongje Seong, Nayeon Kim, Sujin Jang, Daehyun Ji *filed on 11/04/2024*
- Method and Apparatus with Vector Map Learning and Generation** *US-Application No. 18/605,119*
Nayeon Kim, Sujin Jang, Dae Ung JO *filed on 03/14/2024*
- Method and Apparatus with Lane Generation** *US-Application No. 17/862,821*
Nayeon Kim, Moonsub Byeon, Dokwan Oh, Daehyun Ji *filed on 07/12/2022*
- Method and Apparatus for Determining Slope of Road Using Side View Camera of Vehicle** *US-Application No. 17/685,917*
Moonsub Byeon, Nayeon Kim, Daehyun Ji *filed on 03/03/2022*

Awards & Honors

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- Employee of the Year** *Samsung Electronics 2025*
- CEO Commendation Award ('23.4Q DS Vision Awards)** *Samsung Electronics 2024*
- Business Division Commendation Award (1H 2022)** *Samsung Electronics 2022*
- Grand Award (1st Place), Cloud Programming World Cup (CPWC)** *FORUM8 2017*
- Kookmin University Academic Excellence Scholarship** *KMU 2015 - 2017*

Projects

- Automated Semiconductor Cross-Section Analysis** *Samsung Electronics Jan 2025 – Present*
- Developed an automated system using generative models to analyze semiconductor cross-sectional images and automatically determine optimal stopping points during the milling process.
- Large Language Model Alignment Tuning** *Samsung Electronics June 2024 – Dec 2024*
- Conducted research on alignment tuning methods to enhance LLM performance and safety for internal model development.
- Online Vectorized HD Map Construction** *Samsung Electronics Jan 2023 – May 2024*
- Developed an algorithm of vectorized HD map construction that can predict road information in challenging driving road scenes.
 - Developed an algorithm that effectively constructs vectorized HD map using only a camera, by applying knowledge distillation from the multi-modal fusion of lidar and camera.
- Lane Detection** *Samsung Electronics Sep 2019 – Dec 2022*
- Developed an algorithm to predict 3-dimension lane information from a single image that is robust to various driving environments.
 - Developed a 2-dimension lane fitting algorithm that can be implemented in real vehicles.
- Unsupervised Depth Prediction** *KETI Dec 2017 – Oct 2018*
- Developed an algorithm to improve the performance of the depth map by predicting the movement between images through the flow network.
- International Student Car Competition** *KMU Jan 2018 – May 2018*
- Developed a real-time lane detection algorithm for a 1/4 scale autonomous vehicle, validated through simulation in real-road environments.
- Embedded Software Contest** *KMU June 2017 – Nov 2017*
- Developed embedded software that can be implemented in mini cars to enable autonomous driving on the track.
- GM PACE Collaboration and Innovation Challenge** *KMU Jan 2017 – July 2017*
- Developed multi-level driving automation system that predicts driver behavior via an eye-tracker, validated in both a driving simulator and a real-world vehicle.

Academic Service

Teaching Assistant: Introduction to Engineering Design *Sep 2017 - Dec 2017*
Conference Reviewer: NeurIPS *2025*

Skills

Programming Languages: Python, C++, C
Frameworks: PyTorch, ROS, OpenCV
Tools: Linux, Git, Docker
Languages: Korean(Native), English